

# Nifty 100 Analytics & Valuation Suite

## Analyst Guide

A practical guide to navigating the institutional equity analytics dashboard, understanding its screens, and interpreting the underlying data structure.

### 1. System Overview

The Nifty 100 Analytics & Valuation Suite is an institutional equity analytics dashboard and valuation engine covering 92 constituents of the Nifty 100 index. The project is built with Python, Streamlit, Plotly, and SQLite.

### 2. Running the Dashboard

**Local:** Run `streamlit run src/dashboard/app.py` from the project root.

**Google Colab:** The documented workflow starts Streamlit on port 8501 in headless mode and exposes it using Cloudflare Tunnel.

### 3. Dashboard Screens

| Screen                    | Purpose                                     | Key analyst views                                                                       |
|---------------------------|---------------------------------------------|-----------------------------------------------------------------------------------------|
| 1. Market Overview        | Aggregate pulse of the index                | 6 KPI cards; 11-sector donut; Top 5 Quality Compounders; FY 2019–2024 selector          |
| 2. Company Profile        | Single-company fundamental evaluation       | Company metadata; ROE, ROCE, margin, D/E, 5Y Revenue CAGR, FCF; historical trends       |
| 3. Parametric Screener    | Multi-variable company filtering            | Quality, Value, Growth, Dividend, Debt-Free, Turnaround presets; 10 sliders; CSV export |
| 4. Peer Comparison        | Relative sector benchmarking                | Sector peer dropdown; 8-axis radar; comparison matrix and benchmark company             |
| 5. Trend Analysis         | Longitudinal metric exploration             | Up to 3 metrics simultaneously; 10-year line charts; YoY annotations                    |
| 6. Sector Analysis        | Sector performance and valuation landslides | Revenue/ROE/cash-flow bubble view; sector median KPI chart                              |
| 7. Capital Allocation Map | Balance-sheet reinvestment categorization   | Heatmap of 92 companies across 8 reinvestment archetypes                                |
| 8. Annual Reports         | Regulatory filing repository                | FY 2019–FY 2024 filing index; report links; availability badges                         |

## 4. Analyst Workflow

| Step | Recommended use                                                                        |
|------|----------------------------------------------------------------------------------------|
| 1    | Start at Market Overview to understand index-level KPIs and sector distribution.       |
| 2    | Use Company Profile to inspect fundamentals and historical trajectory.                 |
| 3    | Use Parametric Screener to narrow the universe using documented presets and filters.   |
| 4    | Use Peer Comparison to place a company in its sector context.                          |
| 5    | Use Trend Analysis to examine longitudinal changes across selected metrics.            |
| 6    | Use Sector Analysis to compare sector-level performance and valuation characteristics. |
| 7    | Use Capital Allocation Map to review reinvestment archetypes across the universe.      |
| 8    | Use Annual Reports to locate filing metadata and available documents.                  |

## 5. Relational Database Schema

The project database is `data/nifty100.db`. The README identifies the following core tables:

| Table            | Role                             | Important fields                                                                  |
|------------------|----------------------------------|-----------------------------------------------------------------------------------|
| companies        | Master directory                 | company_id, company_name, broad_sector, sub_sector, about_company                 |
| financial_ratios | Longitudinal fundamental ratios  | return_on_equity_pct, debt_to_equity, free_cash_flow_cr, interest_coverage, compo |
| pl_statements    | Audited financial records        | sales_revenue, net_profit, operating_profit, pbt, eps                             |
| annual_reports   | Filing metadata and availability | Regulatory URLs and document availability flags                                   |

## 6. Key Metrics Referenced by the Dashboard

- Return on Equity (ROE)
- Return on Capital Employed (ROCE)
- Net / operating margin measures
- Debt-to-Equity (D/E)
- 5-year Revenue CAGR
- Free Cash Flow (FCF)
- Interest Coverage
- Price / Earnings (P/E)
- Price / Book (P/B)
- Composite Quality Score

## 7. Analyst Interpretation Notes

Use the dashboard as an analytical exploration layer: first establish index or sector context, then inspect individual-company fundamentals, relative peers, and historical trends. The Annual Reports screen provides the project's filing/document reference layer for further investigation.

This guide documents the workflow and terminology described by the project README. It does not introduce investment recommendations or conclusions beyond the documented dashboard functionality.